

LINEAR ALGEBRA

1. The number of an integer $n \times n$ matrix with determinant 1 is $c_n = \frac{1}{n+1} \binom{2n}{n}$, $n \geq 1$.
2. If sum of the elements in each column (row) of a square matrix of order n is k , then the sum of the elements in each column (row) of A^m will be k^m .
3. Let A be an $m \times n$ matrix with all entries are distinct. Then the number of sub matrix of order $i \times j$ is $\binom{m}{i} \times \binom{n}{j}$.
4. The number of sub matrix of all order of a $m \times n$ matrix A with all entries are distinct is $\sum_{i=1}^m \sum_{j=1}^n \binom{m}{i} \times \binom{n}{j} = (2^m - 1)(2^n - 1)$.
5. Let A be an $m \times n$ matrix with all entries are equal. Then the number of sub matrix of order $i \times j$ is 1.
6. The number of sub matrix of all order of a $m \times n$ matrix A with all entries are equal is nm .
7. Let V be a finite-dimensional vector space, and let $T : V \rightarrow V$ be linear.
 - (i) If $\text{rank}(T) = \text{rank}(T^2)$, then $R(T) \cap N(T) = \{0\}$ and $V = R(T) \oplus N(T)$.
 - (ii) $V = R(T^k) \oplus N(T^k)$ for some $k > 0$.
8. Let $\Delta_{2n} = \begin{pmatrix} a & 0 & \cdots & 0 & b \\ 0 & a & \cdots & b & 0 \\ \vdots & \vdots & \cdots & \vdots & \vdots \\ b & 0 & \cdots & 0 & a \end{pmatrix}_{2n \times 2n}$. Then $\det(\Delta_{2n}) = (a^2 - b^2)^n$.
9. Let $A \in M_{m \times n}(F)$. Then $\text{rank}(A^*A) = \text{rank}(A)$.
10. If A is an $m \times n$ matrix such that $\text{rank}(A) = n$, then A^*A is invertible.
11. For any square matrix A , $\|A\|$ is finite and, in fact, equals $\sqrt{\lambda}$, where λ is the largest eigenvalue of A^*A .
12. Let A be an invertible matrix. Then $\|A^{-1}\| = \frac{1}{\sqrt{\lambda}}$, where λ is the smallest eigenvalue of A^*A .

13. Let T be a linear operator on V whose characteristic polynomial splits, and let $\lambda_1, \lambda_2, \dots, \lambda_k$ be the distinct eigenvalues of T . Then T is diagonalisable **iff** $\text{rank}(T - \lambda_i I) = \text{rank}(T - \lambda_i I)^2$ for $1 \leq i \leq k$.
14. A linear operator T on a finite-dimensional vector space V is diagonalisable **iff** its Jordan canonical form is a diagonal matrix.
15. The system of linear equations $Ax = b$ has a solution **iff** $b \in R(A)$ (Row space of A).
16. An orthonormal set in an inner product space is linearly independent.
17. Let W_1 and W_2 be subspaces of a finite dimensional inner product space V . Then
 - (i) $(W_1 + W_2)^\perp = W_1^\perp \cap W_2^\perp$
 - (ii) $(W_1 \cap W_2)^\perp = W_1^\perp + W_2^\perp$
18. Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. Then A is diagonalisable **iff**

$$(a + d)^2 > 4(ad - bc).$$
19. For any square matrix $A = [a_{ij}]$ of order n ,

$$R(A) = \max\{R_i(A) = \sum_{j=1}^n |a_{ij}| : 1 \leq i \leq n\}$$

$$C(A) = \max\{C_j(A) = \sum_{i=1}^n |a_{ij}| : 1 \leq j \leq n\}$$

$$s_i = R_i(A) - |a_{ii}|$$
20. For any square matrix A of order n , every eigen value λ of A satisfies $|\lambda - a_{ll}| \leq s_l$ for some $1 \leq l \leq n$.
21. For any square matrix A of order n , every eigen value of A satisfies $|\lambda| \leq \min\{R(A), C(A)\}$.

22. The eigen value of the tridiagonal matrix $A = \begin{bmatrix} a & b_1 & 0 \cdots & \cdots & 0 \\ c_1 & a & b_2 & \cdots & \\ & c_2 & a & b_3 & \\ \vdots & \cdots & \cdots & a & b_{n-1} \\ & \cdots & & c_{n-1} & a \end{bmatrix}_{n \times n}$

$$(n \geq 3)$$
 satisfies the inequality $|\lambda - a| < 2\sqrt{\max_i |b_i| \max_i |c_i|}$.

23. If A is idempotent. then $\text{Rank}(A) = \text{Trac}(A)$.
24. If A is symmetric matrix, then $\text{Rank}(A)$ is equal to the number of non-zero eigen values of A .

25. Test for diagonalization

Let T be a linear operator on an n -dimensional vector space V . Then T is diagonalizable **iff**

- (i) The characteristic polynomial of T splits.
- (ii) For each eigen value λ of T , the multiplicity of λ equals $n - \text{rank}(T - \lambda I)$.

- 26.

$$\begin{aligned} \text{rank} \left(\begin{bmatrix} A & B \\ C & D \end{bmatrix} \right) &= \text{rank}(A) + \text{rank}(D - BA^{-1}C) \\ &= \text{rank}(D) + \text{rank}(A - CD^{-1}B). \end{aligned}$$

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$$\begin{aligned} \det \left(\begin{bmatrix} A & B \\ C & D \end{bmatrix} \right) &= \det(A)\det(D - BA^{-1}C) \\ &= \det(D)\det(A - CD^{-1}B). \end{aligned}$$

28. Let T be a linear operator on a finite dimensional vector space over $\mathbb{C}$. Then T is diagonalisable **iff** T is annihilated by some polynomial f over $\mathbb{C}$ which has distinct roots.
29. A linear operator T on a finite-dimensional vector space V is diagonalizable **iff** V is the direct sum of the eigen spaces of T .
30. Any projection E on a vector space V is diagonalizable.
31. If N is nilpotent, then $I + N$ is invertible.
32. If N is nilpotent, then $\det(I + N) = 1$.
33. If A is a matrix and $\det(I + tA) = 1$ for all t , then A is nilpotent.
34. Every singular matrix can be written as a product of nilpotent matrices.

35. For an $n \times n$ square matrix N with real (or complex) entries, the following are equivalent:
- (i) N is nilpotent
 - (ii) The minimal polynomial of N is λ^k for some positive integer $k \leq n$
 - (iii) The characteristic polynomial for N is λ^n
 - (iv) The only eigen values of N is 0
 - (v) $\text{trace}(N^k) = 0$ for all $k > 0$.
36. Dimension of the vector space of $n \times n$ matrices when trace is zero is $n^2 - 1$.
37. If $V = \{A = (a_{ij})_{m \times n} : \sum_{j=1}^n a_{ij} = 0, \forall i = 1, 2, \dots, m\}$
- (i) When column is zero then dimension of V is $mn - m$
 - (ii) If one row and one column is zero, then its dimension is $(m - 1)(n - 1) = mn - m - n + 1$
38. The dimension of symmetric matrices of vector space is $\sum n = \frac{n(n+1)}{2}$
39. The dimension of skew symmetric matrices is $\sum (n - 1) = \frac{n(n-1)}{2}$
40. The dimension of symmetric matrices with trace zero is $\frac{n(n+1)}{2} - 1$
41. The dimension of skew-symmetric matrices with trace zero is $\frac{n(n-1)}{2}$
42. Vector space of $n \times n$ matrix with complex field over the real number have dimension $2n^2$.
43. Dimension of $n \times n$ Hermitian matrices with complex entry over the field of real numbers is $2n^2$.
44. Vector space of $n \times n$ skew-Hermitian matrices over $\mathbb{R}$, then it's dimension is n^2 .
45. Vector space of upper triangular matrices, it's dimension is $\sum n$
46. Vector space of lower triangular matrices, it's dimension is $\sum n$

47. The dimension of scalar matrices is 1.
48. Dimension of diagonal matrices is n .
49. Dimension of scalar matrices and trace zero is 0.
50. The number of basis in the finite dimensional n vector space over a finite field $\mathbb{Z}_p$ is,

$$\frac{(p^n - 1)(p^n - p) \cdots (p^n - p^{n-1})}{n!}$$

51. The number of subspaces of dimension r is

$$\frac{(p^n - 1)(p^n - p) \cdots (p^n - p^{r-1})}{(p^r - 1)(p^r - p) \cdots (p^r - p^{r-1})}$$

52. $|adj A| = |A|^{n-1}$, if $|A| \neq 0$
53. $|adj \cdot adj \cdots adj A| = |A|^{(n-1)r}$
54. (i) $rank(A) = n \Rightarrow rank(adj A) = n$
 (ii) $rank(A) = n - 1 \Rightarrow rank(adj A) = 1$
 (iii) $rank(A) < n - 1 \Rightarrow rank(adj A) = 0$
 (iv) $rank(AB) \leq \min\{rank(A), rank(B)\}$
55. Let A and B be any $n \times n$ matrix.
 Then $rank(A) + rank(B) - n \leq rank(AB) \leq \min\{rank(A), rank(B)\}$.
56. $rank(A + B) \leq rank(A) + rank(B)$.
57. Let A be an $m \times n$ matrix and B be any $n \times p$ matrix. Then
 Then $rank(AB) \geq rank(A) + rank(B) - n$.
58. $|A - \lambda I| = \lambda^3 - trace(A)\lambda^2 + trace(adj(A))\lambda - |A| = 0$.
59. Sum of the eigen values = $trace(A)$.
60. Product of eigen values = $\det(A)$.
61. If A is nilpotent then $A + \lambda I$ is always non-singular for any value of λ .
62. The eigen values of symmetric / Hermitian matrix are real.

63. The eigen values of skew-Hermitian is purely imaginary.
64. The skew-Hermitian matrices have even rank.
65. The determinant of skew-Hermitian matrix of odd order is zero.
66. For a 2×2 matrix, the eigen values are
- $$\lambda = \frac{[(a_{11} + a_{22}) \pm \sqrt{4a_{12}a_{21} + (a_{11} - a_{22})^2}]}{2}$$
67. If $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by
- $$T(x, y) = (ax + by, cx + dy)$$
- is $\mathbb{C}$ -linear **iff** $a = d$ and $b = -c$.
68. Let A be an $n \times n$ matrix (a_{ij}) with complex entries. Suppose $|a_{ii}| > \sum_{j=1}^n |a_{ij}|$, $j \neq i$, for all $i = 1, 2, \dots, n$. Then A is invertible.
69. Let A be an $m \times n$ matrix and $W_A = \{x \in \mathbb{R}^m \mid xA = 0\}$. Then $\dim W_A = m - \text{rank}(A)$.
70. Let A be an $m \times n$ matrix and B be an $n \times p$ matrix and $W_{AB} = \{x \in \mathbb{R}^m \mid xAB = 0\}$. Then $\dim W_{AB} = m - \text{rank}(AB)$.